## Лаб №15

**Зорилго: Django Class Based View. CreateView, UpdateView, DeleteView ашиглаж сурах.**

Өмнөх лаб дээр үргэлжлүүлэн хийж гүйцэтгэх бөгөөд өөрчлөлт орсон кодыг Bold-р тэмдэглэв.

1. Django-project үүсгэнэ.

```
django-admin startproject ComedyImage
```

2. Project дотор app үүсгэнэ.

```
Django-admin startapp basic_app
```

3. Templates folder үүсгэх тохируулах base.html болон index.html 2 хуудас нэмж өгнө.

**### templates/basic_app/base.html код ###**

```
<html>
    <head>
        <meta charset="utf-8">
        <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
        <title>Base</title>
    </head>
    <body>
        <nav class="navbar navbar-default navbar-static-top">
            <ul class="nav navbar-nav">
                <li><a class="navbar-link" href="{% url 'basic_app:list' %}"
>Комеди клубууд</a></li>
                <li><a  class="navbar-link"  href="{%  url  'admin:index'  %}"
>Админ</a></li>
                <!--<li><a class="navbar-brand" href="{% url 'basic_app:list'
%}" >Schools</a></li>-->
            </ul>
        </nav>
        <div class="container">
            {% block body_block %}
            {% endblock %}
        </div>
    </body>
</html>
```

Index.html

**### templates/basic_app/index.html код ###**

```
{% extends "base.html" %}
{% block body_block %}
    <div class="jumbotron">
```

```
        <h1>Монголын комеди клубууд вэб апп.</h1>
    </div>
{% endblock %}
```

4. Static folder үүсгэж тохируулах. Static folder дотор image нэртэй дэд folder үүсгэж өгнө. (Лаб №7 харна уу.)

```
STATIC_DIR = os.path.join(BASE_DIR, "static")
STATICFILES_DIRS = [
    STATIC_DIR,
]
```

5. Model үүсгэх. Club болон Comedian

```
### basic_app/models.py код ###
from django.db import models

# Create your models here.

class Club(models.Model):
    name = models.CharField(max_length=256)
    location  = models.CharField(max_length=256)

    def __str__(self):
        return self.name

class Comedian(models.Model):
    name = models.CharField(max_length=256)
    age = models.PositiveIntegerField()
    image = models.ImageField(blank=True, null=True,
                              upload_to='static/images')
    club = models.ForeignKey(Club, related_name='clubs',
                              on_delete=models.CASCADE)

    def __str__(self):
        return self.name
```

6. Өгөгдлийн сан үүсгэх, superuser үүсгэх, model-уудыг бүртгүүлэх.

```
Python manage.py migrate
Python manage.py makemigrations
Python manage.py createsuperuser
```

```
### basic_app/admin.py код ###
from django.contrib import admin
from basic_app.models import Club, Comedian

admin.site.register(Club)
admin.site.register(Comedian)
```

7. Admin хуудас ашиглан comedy club болон тухайн club-д хамаарах comedian-уудыг нэмж өгнө.
8. View үүсгэх

```
### basic_app/views.py код ###
from django.shortcuts import render
from django.views.generic import (View, TemplateView, ListView,
                                  DetailView, CreateView,
                                  UpdateView, DeleteView)
from . import models
from django.urls import reverse_lazy
# Create your views here.

class IndexView(TemplateView):
    template_name = 'index.html'

class ClubListView(ListView):
    context_object_name='clubList'
    model = models.Club

class ClubDetailView(DetailView):
    context_object_name='clubDetails'
    model = models.Club
    template_name = 'basic_app/club_detail.html'

class ClubCreateView(CreateView):
    fields = ('name', 'location')
    model = models.Club

class ClubUpdateView(UpdateView):
    fields = ('name', 'location')
    model = models.Club

class ClubDeleteView(DeleteView):
    model = models.Club
    success_url = reverse_lazy("basic_app:list")
```

9. Basic_app дотор templates folder үүсгэх бөгөөд basic_app нэртэй дэд folder үүсгэж өгнө. Basic_app → templates → basic_app-д basic_app_base.html, club_list.html, club_detail.html үүсгэнэ.

```
### basic_app/templates/basic_app/basic_app_base.html код ###
```

```
<html>
    <head>
        <meta charset="utf-8">
        <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
        <title>Base</title>
    </head>
    <body>
        <nav class="navbar navbar-default navbar-static-top">
            <ul class="nav navbar-nav">
                <li><a class="navbar-link" href="{% url 'basic_app:list' %}"
>Комеди клубууд</a></li>
                <li><a class="navbar-link" href="{% url 'admin:index' %}"
>Админ</a></li>
                </ul>
        </nav>
        <div class="container">
            {% block body_block %}
            {% endblock %}
        </div>
    </body>
</html>
```

Club_list.html

```
### basic_app/templates/basic_app/club_list.html код ###
```

```
{% extends "basic_app/basic_app_base.html" %}

{% block body_block %}

<h1>Комеди клубууд!</h1>
    <ol>
        {% for club in clubList %}
        <h2><li><a href="{{ club.id }}"> {{ club.name }}</a></li></h2>
        {% endfor %}
    </ol>

{% endblock %}
```

Club_details.html

```
### basic_app/templates/basic_app/club_details.html код ###
```

```
{% extends "basic_app/basic_app_base.html" %}
{% block body_block %}
<div class="jumbotron">
    <h2>Клубын талаар дэлгэрэнгүй</h2>
    <p>Нэр: {{ clubDetails.name }}</p>
    <p>Хаяг: {{ clubDetails.location }}</p>
    <h3>Жүжигчид:</h3>

    {% for comedian in clubDetails.clubs.all %}
        <p>Нэр:{{ comedian.name }}.</p>
        <p><img src="../../{{comedian.image}}" width="100"></p>
    {% endfor %}
</div>
<div class="container">
    <p><a class="btn btn-warning" href="{% url 'basic_app:update'
pk=clubDetails.pk %}">Засах</a></p>
</div>
{% endblock %}
```

10. Basic_app urls.py файл үүсгэх.

```
### basic_app/urls.py код ###
from django.urls import path
from basic_app import views

app_name = 'basic_app'

urlpatterns = [
    path('', views.ClubListView.as_view(), name='list'),
    path('<pk>/', views.ClubDetailView.as_view(), name='detail'),
    path('create/', views.ClubCreateView.as_view(), name='create'),
    path('update/<pk>/', views.ClubUpdateView.as_view(), name='update'),
    path('delete/<pk>/', views.ClubDeleteView.as_view(), name='delete'),
]
```

11. ComedyImage project urls.py тохируулах.

```
### ComedyImage/urls.py код ###
from django.contrib import admin
from django.urls import path, include
from basic_app import views
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', views.IndexView.as_view()),
    path('basic_app/',  include('basic_app.urls', namespace='basic_app')),
]
```

12. Лаб №15 дээр нэмэгдэх template-үүд ComedyImage → basic_app → templates → basic_app

- club_form.html
- club_confirm_delete.html

```
### basic_app/templates/basic_app/club_form.html код ###
{% extends "basic_app/basic_app_base.html" %}

{% block body_block %}

<h1>
    {% if not form.instance.pk %}
    Create Club
    {% else %}
    Upadate Club
    {% endif %}
</h1>
    <div class="jumbotron">
    <form method="POST">
        {% csrf_token %}
        {{ form.as_p }}
        <input type="submit" class="btn btn-primary" value="Submit">
    </form>
    </div>
{% endblock %}
```

```
### basic_app/templates/basic_app/club_confirm_delete.html код ###
{% extends "basic_app/basic_app_base.html" %}
{% block body_block %}

<h1>Delete {{ club.name }}?</h1>
<form method="POST">
    {% csrf_token %}
    <input type="submit" class="btn btn-danger" value="Delete">
    <a href="{% url 'basic_app:detail' pk=club.pk %}">Cancel</a>
</form>

{% endblock %}
```